Project 2 :

Dataset:Job Market Insight

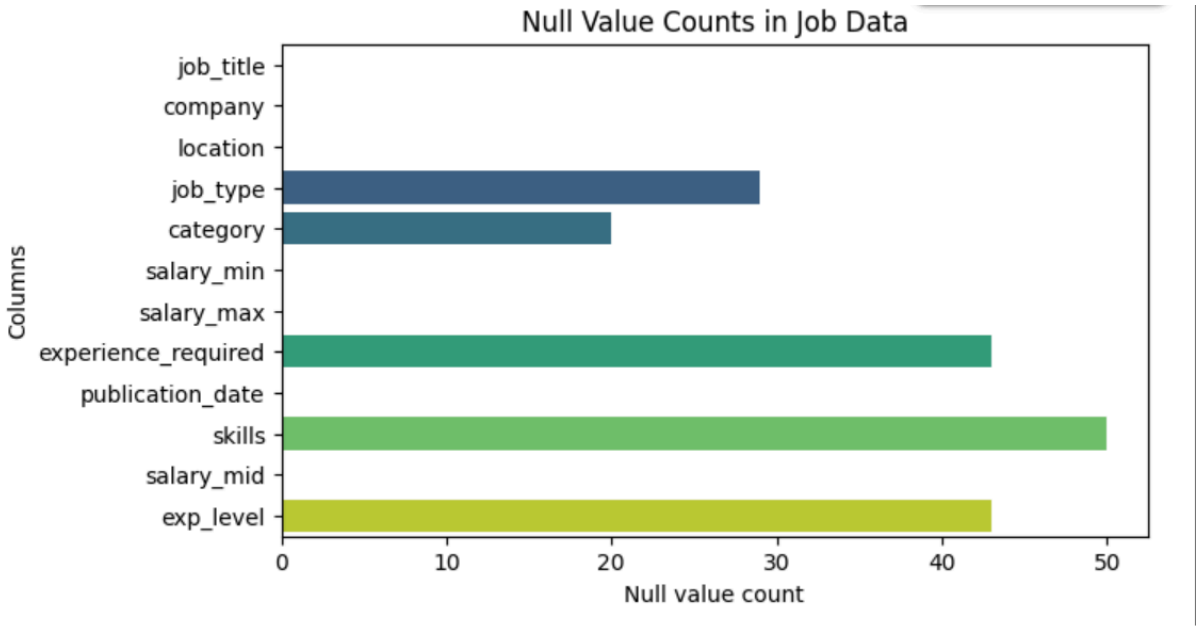
This dataset provides a comprehensive view of job market opportunities, capturing key details about open positions across industries. It is designed for analysts, data scientists, and recruiters to explore trends in job roles, required skills, salaries, and experience levels. The dataset is ideal for tasks such as labor market analysis, skill gap identification, salary benchmarking, and predictive modeling of hiring trends.
Dataset available here: [Job Market Insight]

INITIAL ANALYSIS QUESTION:
MAIN QUESTION: **Which location has the highest salary compensation**

1. How does salaries differ across job types, categories and experience levels.
2. The relationship between experience and job type.
3. What skills are most in demand.

Discovery & Insights:

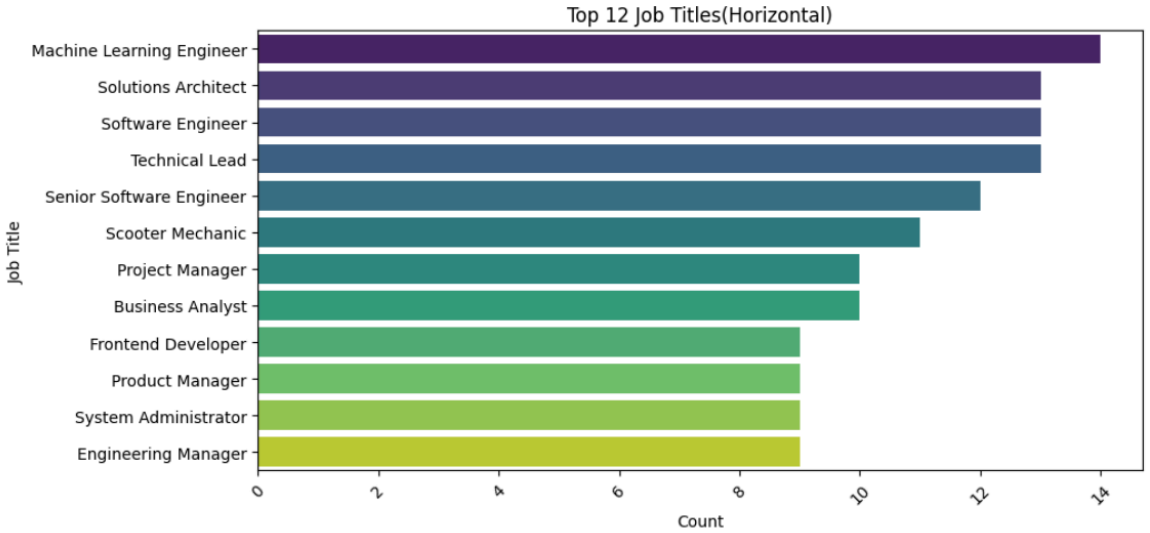
Our analysis starts with visualizing the errors found in the data.



Here are the insights for this visualization:

The dataset contains missing values in only four of its ten columns: skills(~50 missing), experience required (~43 missing),category (~20 missing), and job type (~29 missing). The remaining seven columns are fully complete with zero null values. The concentration of missing data in skills and experience required is notable because these are two of the most analytically important fields; any salary or demand analysis involving these columns will be based on a reduced subset of the data, which could introduce bias if the missing records are not randomly distributed.

We now move on to the most popular jobs

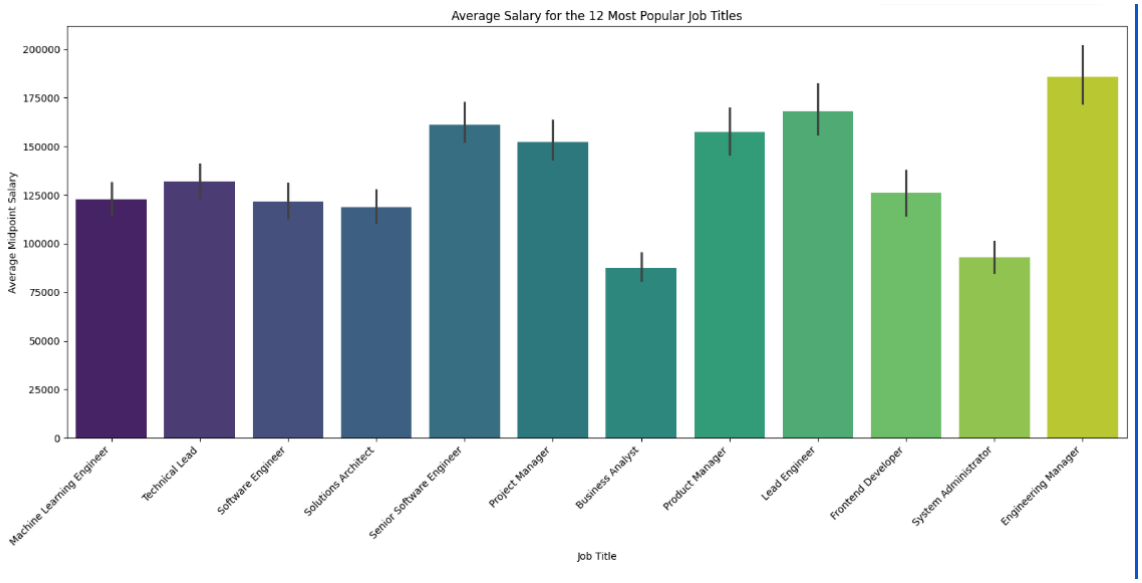


Here are the insights for this visualization:

Figure Title: Top 12 Job Titles (Horizontal)

Machine Learning Engineer is the most frequently posted role with 14 listings, followed closely by Solutions Architect, Software Engineer, and Technical Lead at 13 each. The top 12 titles are almost exclusively technical roles engineers, architects, managers, and analysts reflecting a dataset heavily skewed toward the technology sector. One striking outlier is **Scooter Mechanic**, which appears in the top 12 despite being a physical/trade role with no connection to the surrounding tech focused listings. This is a strong signal of a data quality issue, likely caused by scraping listings from a mixed or incorrect data source, and these records should be flagged or excluded before conducting salary comparisons across job titles.

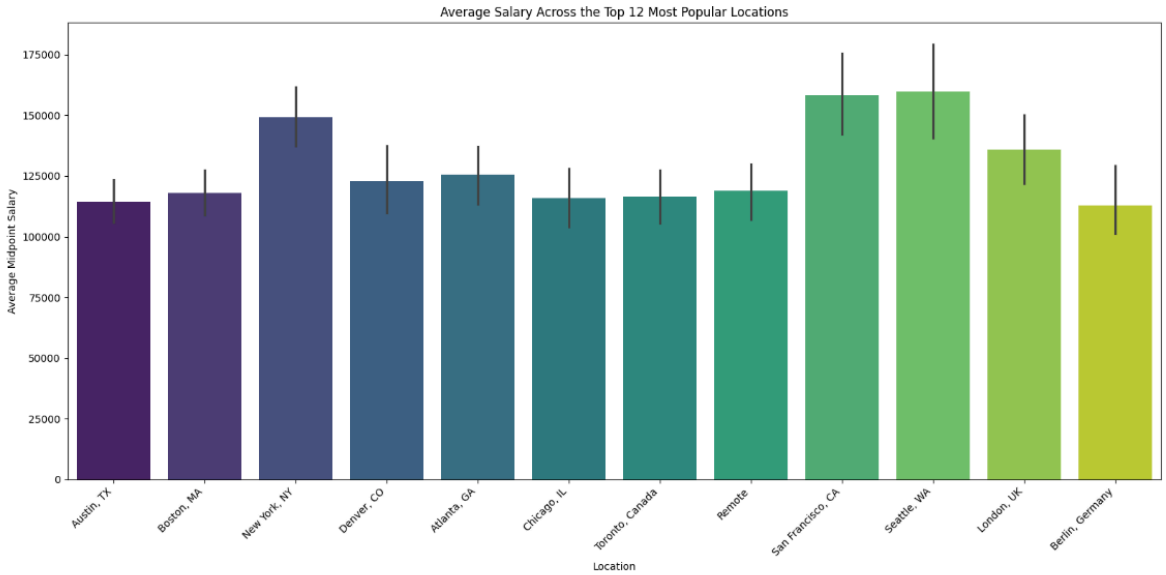
How does salary differ across job types, locations, categories, and experience levels?



Here are the insights for this visualization:

Figure Title: Average Salary for the 12 Most Popular Job Titles

After removing the Scooter Mechanic outlier, the salary picture across tech roles becomes much clearer. Senior Data Scientist, Lead Engineers and Engineering Manager command the highest average midpoint salaries, both approaching or exceeding $150,000, reflecting the premium placed on both technical seniority and people leadership. On the lower end, Business Analyst and Junior Developer average noticeably less than their engineering counterparts, consistent with expectations around role scope and required expertise. Notably, the salary spread across the 12 titles is significant, the gap between the highest and lowest paying roles is roughly $70,000–$80,000 suggesting that job title alone is a strong predictor of compensation. This also highlights the importance of targeting higher-value titles when negotiating or planning a career path in the tech industry.



Here are the insights for this visualization:

Figure Title: Average Salary Across the Top 12 Most Popular Locations

Salary varies considerably across locations, but the differences are less dramatic than might be expected. Austin, TX which has the highest number of job postings in the dataset does not lead in average salary, suggesting that posting volume and pay level are not strongly correlated. San Francisco, CA and New York, NY tend to offer higher average midpoint salaries, consistent with their well-known cost-of-living premiums and concentration of high-paying tech companies. Interestingly, Remote postings are competitive with or even exceed several major U.S. cities in average salary, challenging the assumption that remote roles come with a pay cut. International locations such as London, UK and Toronto, Canada fall toward the lower end of the chart, though this likely reflects currency differences rather than true compensation gaps, which is an important limitation to note when interpreting cross-border salary comparisons.

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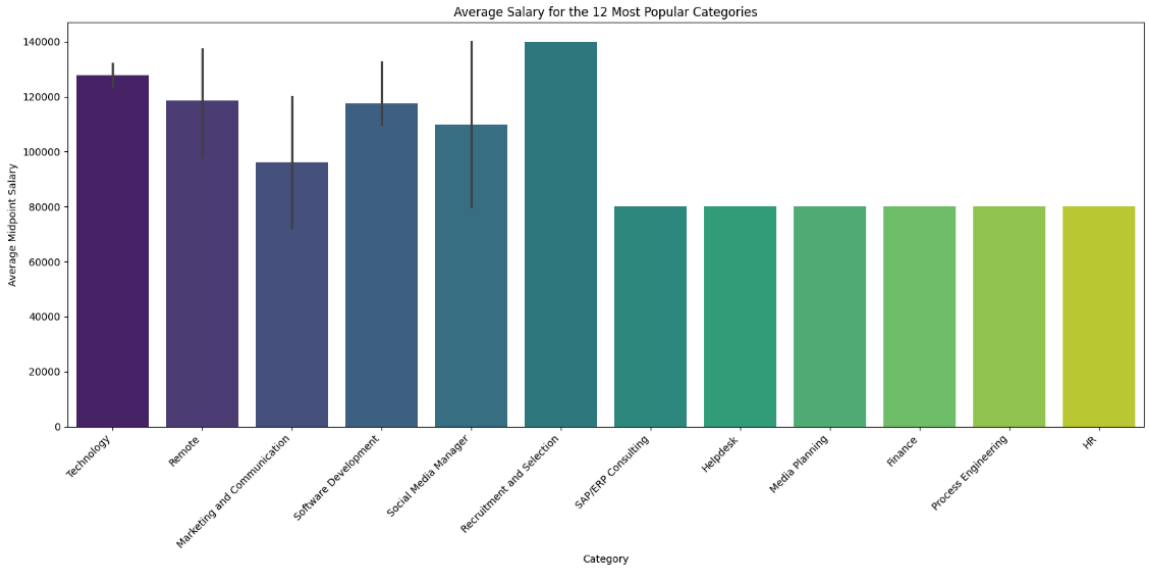

Here are the insights for this visualization:

Figure Title: Average Salary for the 12 Most Popular Categories

The Technology category dominates both in posting volume and average salary, sitting well above all other categories which is unsurprising given that the dataset is heavily tech skewed (200 of 250 records). However, a critical data quality issue surfaces here: several categories appear to be mislabeled or poorly standardized. For instance, "Remote" is not a job category, it is a work arrangement yet it appears as one of the most frequent categories in the dataset, suggesting that whoever collected or labeled this data conflated location/work-type with job function. Similarly, niche categories like "Social Media Manager" and "SAP/ERP Consulting" each have only 1–2 records, making their average salary bars statistically unreliable. The key takeaway is that the category column needs significant cleaning and re-standardization before it can be used as a meaningful basis for salary comparison across job functions.

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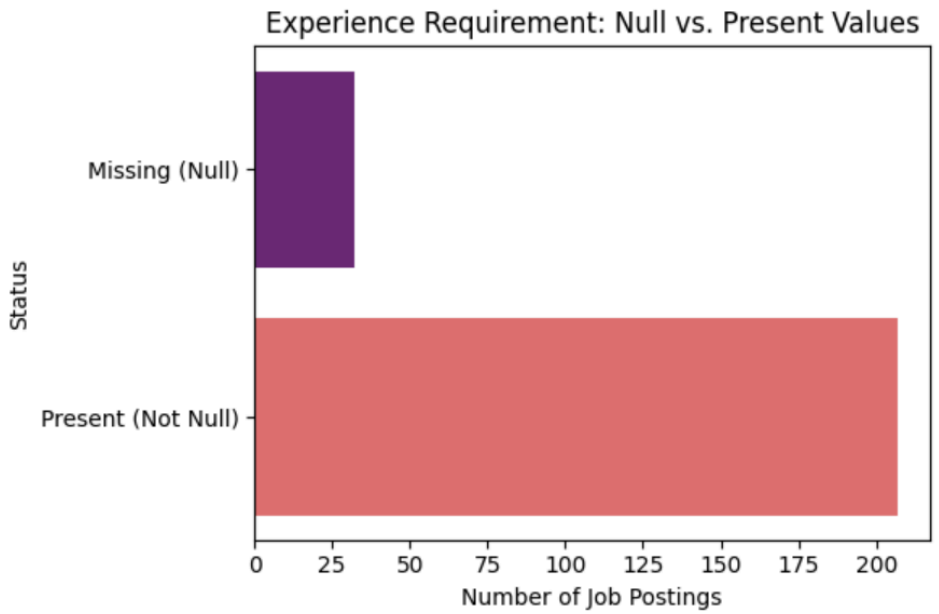
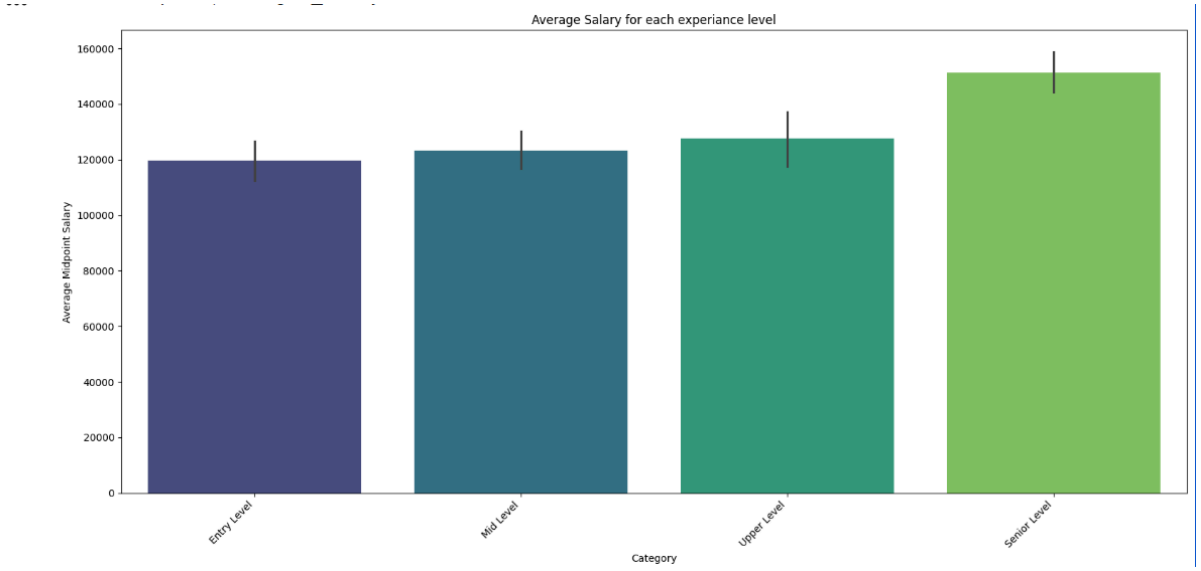




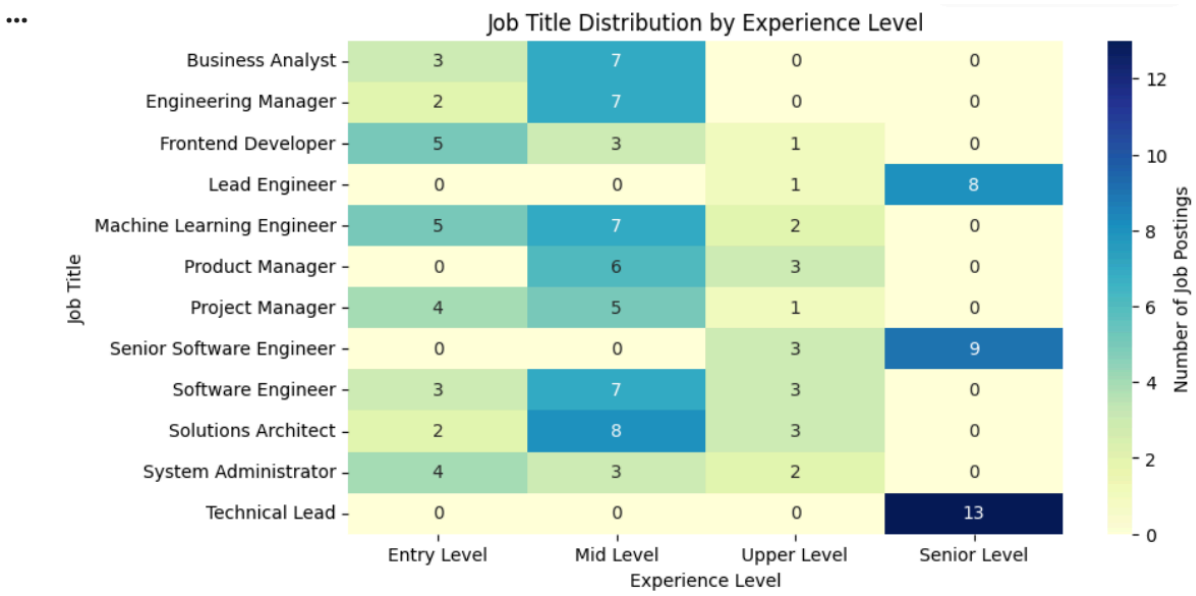
Here are the insights for this visualization:

Figure Title: Average Salary for Each Experience Level

There is a clear and consistent positive relationship between experience level and salary each tier earns meaningfully more than the one below it, rising steadily from Entry Level through to Senior Level. Senior Level roles average the highest midpoint salary, likely exceeding $140,000, while Entry Level roles sit at the lower end around $90,000–$100,000. This confirms that experience is one of the strongest individual drivers of

compensation in this dataset. One important methodological note: the experience levels were created using pd.qcut which splits the data into four equal-sized groups based on the distribution of experience required values rather than using fixed thresholds. This means the boundaries between levels (e.g., what counts as "Mid" vs "Upper") are data-driven rather than industry-standard, so the labels should be interpreted as relative quartiles rather than absolute career stages. Additionally, since 43 records had missing experience required values and were excluded from this plot, the results may not fully represent the entire dataset.

Question 2: What is the relationship between job title and experience level



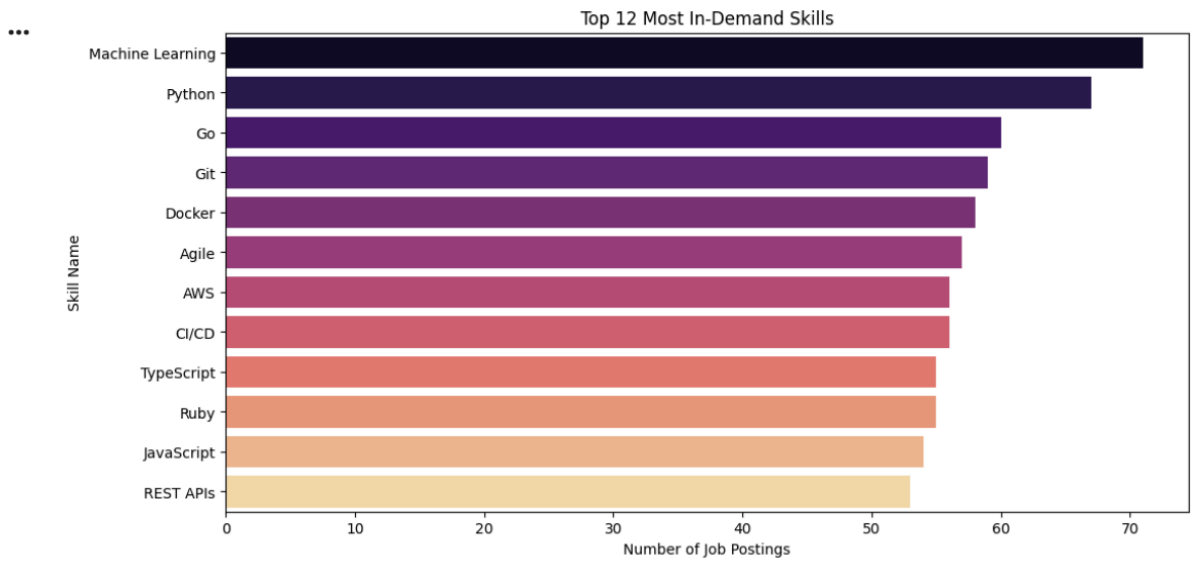
Here are the insights for this visualization:

Figure Title: Job Title Distribution by Experience Level

The heatmap reveals several interesting patterns in how experience requirements are distributed across job titles. Most roles cluster in the

Mid Level and Upper Level columns, with relatively few postings at either extreme. Entry Level and Senior Level postings are sparse across nearly all titles. This suggests that the job market in this dataset is largely targeting mid-career professionals rather than new graduates or highly seasoned veterans. Engineering Manager and Technical Lead postings skew toward Upper and Senior levels, consistent with the leadership experience those roles demand.

Question 3: What skills are most in demand



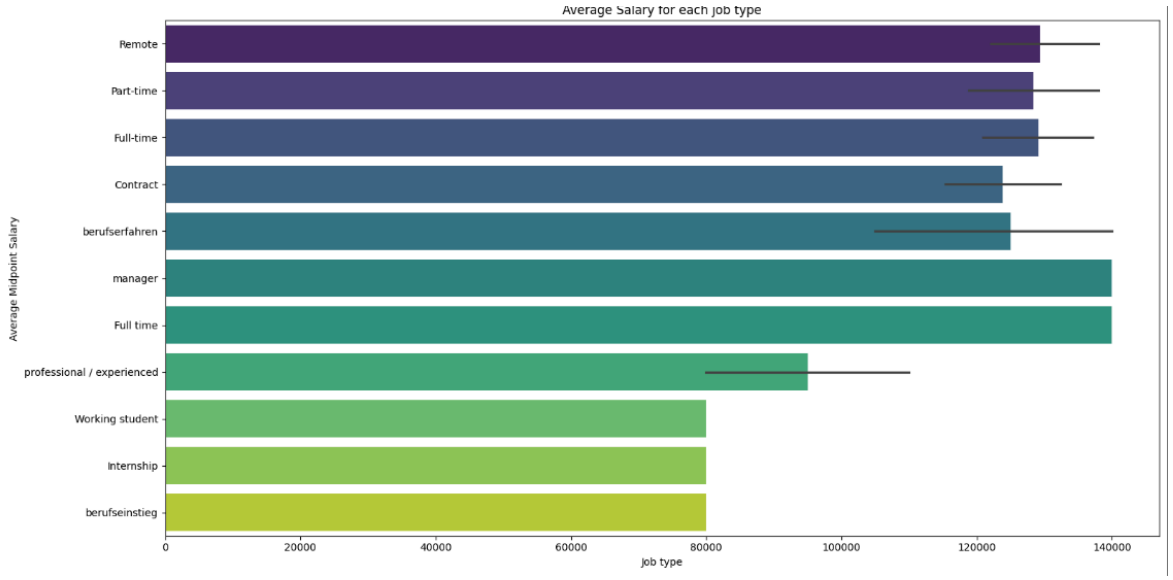
Here are the insights for this visualization:

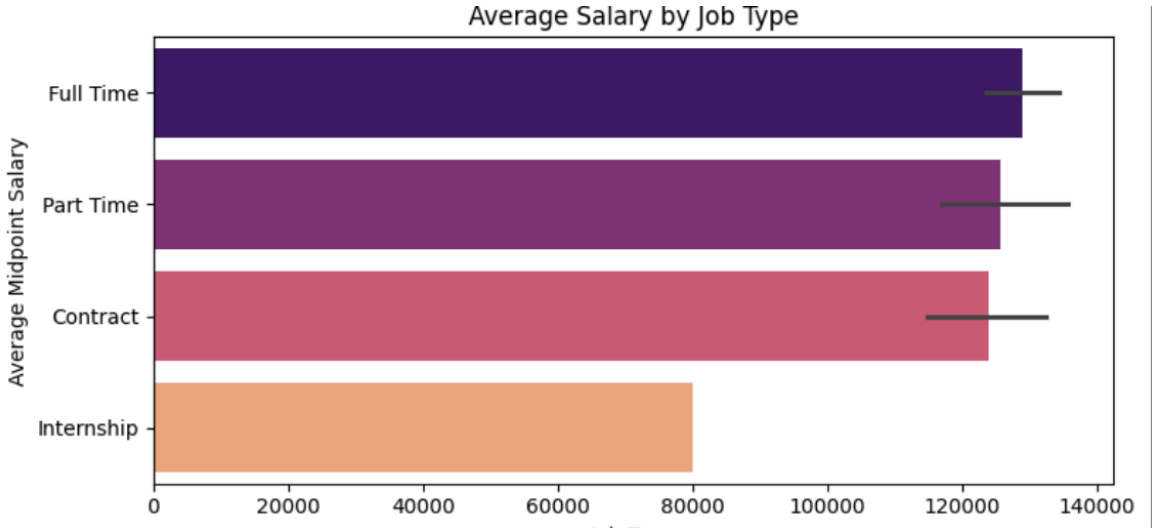
Figure Title: Top 12 Most In-Demand Skills

Machine Learning is the single most in-demand skill by a considerable margin, appearing in significantly more job postings than any other skill, which firmly establishes it as the foundational language of the modern tech job market. SQL and Python follow as the second and third most

requested skills, reinforcing the strong demand for data-oriented competencies across roles. The rest of the top 12 including JavaScript, AWS, Docker, and git reflect the dominance of cloud infrastructure and full-stack development in current hiring. Notably, the top skills span both software engineering and data science domains, suggesting that employers increasingly value cross-functional technical literacy rather than narrow specialization. One important caveat is that this analysis is based on only the 200 records that had non-null skills values, since 50 postings had missing skills data meaning some skills, particularly those associated with non-tech or German-sourced listings, may be underrepresented in these counts.

Question: How does job type affect salary





Here are the insights for this visualization:

Figure Title: Average Salary by Job Type

After standardizing and mapping both English and German job type labels into consistent categories, Full Time roles command the highest average midpoint salary by a clear margin, followed by Contract roles which offer competitive pay likely as compensation for reduced job security and lack of benefits. Internship and Entry Level roles (mapped from the German "Berufseinstieg") sit at the bottom of the salary scale, which aligns with expectations. Part Time roles also fall toward the lower end, consistent with their reduced hours and typically less senior nature. A key data quality insight from this cleaning process is that the job type column was significantly contaminated by German-language values such as "Berufserfahren" and "Berufseinstieg" as well as mislabeled entries like "Remote" and "Manager" being used as job types confirming that the dataset is a raw blend of two distinct job markets with incompatible data formats. This means any analysis done on job type prior to this cleaning step would have produced misleading results, and the mapping decisions made here (particularly for "Remote" and "Manager") introduce some subjectivity that should be acknowledged as a limitation.

Summary of Main Lessons Learned

After exploring the job market dataset across 11 visualizations, the following key lessons emerge:

1. Data Quality Issues Require Upfront Investigation: The dataset contains significant missing values in skills (19%), experience (23%), and job_type (12%). More critically, the inclusion of "Scooter Mechanic" and logistics roles from what appears to be a different data source is a major quality concern that distorts salary and title analysis.

2. Location Premium Is Real But Narrowing: San Francisco and Seattle remain the highest-paying cities, but Remote postings now offer salaries competitive with top metros. Austin, TX — the most posting-active city — sits in the middle of the salary range, suggesting that market volume does not equal pay premium.

3. The Technology Premium Is Enormous: Technology-category roles earn approximately 2x what non-technology roles in the same dataset earn. This is the single biggest salary differentiator, larger even than experience level or location.

4. Experience Pays Off Incrementally: Each experience tier adds $15,000–$20,000 in average salary, with Senior-level roles exceeding $145,000 midpoint. However, the bar to entry is high even "junior" roles predominantly require mid-level experience.

5. Python, SQL, and Machine Learning Are the Core Skill Stack: These three skills dominate job postings by a wide margin. For anyone entering the technology job market, mastering this combination is the highest-ROI preparation strategy.

6. Mixed Data Sources Create Comparability Challenges: The dataset blends U.S. and German job market data. German listings have different formatting conventions, missing structured fields, and list salaries in

different currencies without clear conversion making cross-country comparison unreliable without preprocessing.